

SOUMYA GHORAI

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🎓 Google Scholar 🌐 Personal Website

Education

Indian Institute of Technology Kharagpur, India

Sept 2022 – Present

Ph.D. Candidate, Electrical Engineering

Advisor: Souvik Chattopadhyay

Committee Members: Debaprasad Kastha, Santanu Kapat, Pradip Mandal

Funding: *Prime Minister Research Fellowship*

Jadavpur University, Kolkata, India

2017 – 2019

M.Tech., Energy Science & Technology

Thesis: Sectoral harmonization of electric vehicle transport and power approaching demand side energy management: a case study in West Bengal

Thesis Advisor: Tushar Jash

Jadavpur University, Kolkata, India

2013 – 2017

B.E., Electrical Engineering

Research Interest

My research focuses on single-stage AC-DC, DC-AC, and AC-AC bidirectional isolated power electronic circuits that are capable of converting power with high efficiency, active power decoupling, modular power processing, and reduced passive components. Currently, I am focusing on dual-active-bridge (DAB) based single-stage circuits, hardware realization, implementation of control, and different modulation strategies.

Journal Publication

Combined PWM and Phase Shift Control of a Variable DC-Link PFC-Integrated DAB Topology for Wide-Range High-Efficiency EV Chargers

S. Ghorai, A. Chambayil, S. Chattopadhyay

IEEE Journal of Emerging and Selected Topics in Power Electronics (JESTPE), 2025.

Conference Publications from Ph.D. work

Single-Phase Single-Stage Three-Port on-Board Charger with Minimum Hardware Components for EV/PHEV Application

K Ushida, K Nakamura, Y Hotta, S Saha, S Ghorai, S Chattopadhyay

17th Annual IEEE Energy Conversion Conference and Expo (ECCE), 2025

A New Three-phase Single-stage DAB Isolated Phase-modular PFC Topology with Sextuple Active Bridges

S. Ghorai, S. Chattopadhyay

17th IEEE Energy Conversion Conference and Expo (ECCE- Asia), 2025

Dual Active Bridge-Based Isolated Single-phase Single-Stage DC-link AC/AC Converter with Minimum Charge Storage Requirement

S. Ghorai, S. Chattopadhyay

17th IEEE Energy Conversion Conference and Expo (ECCE- Asia), 2025

A Highly Efficient Wide Battery Voltage Range Dual Active Bridge-Based Single-stage Bidirectional EV Onboard Charger Without Low-Frequency Harmonics in the Charging Current
S. Ghorai, S. Chattopadhyay
41st IEEE International Communications Energy Conference (INTELEC), 2024

Invention Dsiclosures

S. Ghorai, S. Chattopadhyay. **A Highly Efficient Active Power Decoupled Sigle-phase Single-stage EV Onboard Charging System With Ripple-less DC Battery Charging Current.** Indian Patent (filed)

S. Ghorai, S. Chattopadhyay. **A Modular Dual-Active Bridge (DAB)- based Three Phase AC-TO-DC Conversion System.** Indian Patent (filed)

S. Ghorai, S. Chattopadhyay. **A Novel Control & Modulation Technique for Single-stage Power Electronic Transformer.** Indian Patent (filed)

Conference Publications from M.Tech. work

PV assisted Fuzzy based EV charge scheduling for demand side energy management: a case study

S. Ghorai, D. Majumdar, T. Jash and S. Ray
2020 IEEE Calcutta Conference (CALCON)

Revisiting small signal stability analysis techniques in the context of stability study of a two-area multimachine power system

S. Ghorai, S. Ray
2020 IEEE Calcutta Conference (CALCON)

Research Experience

Ph.D. Project 1: Development of 50kW Single-stage DAB Isolated Modular DC Fast Charger
CPRI, India -funded Project PI: Dr. Souvik Chattopadhyay *August'23 - present*

- Conducted verification of a novel 3-phase modular single-stage AC-DC DAB topology in *PLECS* platform which consists of six active bridges with negligible DC-link capacitor requirement and enabled with direct line current shaping and minimum RMS current in the high-frequency link.
- Conducted PCB designing in *Altium*, 3D-design of mechanical assembly in *Solidworks*, hardware realization along with design and making the magnetics components, closed-loop-control implementation using Sprtan-6 FPGA board and *VHDL* coding in *XILINX*.
- Currently working towards testing the setup in full power, optimizing the modulation technique to enhance the soft-switching region while ensuring minimum high-frequency rms current during the entire grid cycle, and the ability to work under unbalanced grid voltage conditions.

Ph.D. Project 2: Simulation and Analysis of Universal EV On-Board Battery Charger Configurations with Wide-band gap-based power devices

AISIN CORPORATION, Japan - funded project PI: Souvik Chattopadhyay *Jan'24 - present*

- Conducted hardware realization of a 2.6 kW active power decoupled phase-shift modulated single-phase single-stage AC-DC DAB topology with minimum hardware components. A novel approach of instantaneous phase-shift modulation based on a predictive switching power modulator, is realized on FPGA platform and verified.

Other Projects: Testing of RBC Power Converter

Scientific Power Corporation, Kolkata - funded project PI: Souvik Chattopadhyay *Jan'23 - present*

- Designed a customized control interface board for TI DSP TMS320F28379D for different relays, and display connection (UART communication), multiple PWM and analog sensing interfaces, SD card read and write option via interfacing with the STM 32 blue pill board via SPI communication, gate driver tripping input and output signals.

Past Projects 1: Soft-switched reduced switch high step-down converter

Research Scholar at Indian Institute of Technology Delhi

Aug'20 - March'22

Supervisor: Mummadi Veerachary

- Designed and analyzed a new coupled-inductor-switched-capacitor-cell-based 48V-1V converter topology for VRM applications, which eliminates the leakage inductance caused by the voltage spike problem against the synchronous switch of bucking operation.

Past Projects 2: Off-grid battery-connected PV-fed UV-C LED driver for rural water purifier

Rashtriya Uchchatar Shiksha Abhiyan 2.0 Fellowship, Jadavpur University, Kolkata

June'19 - May'20

PI: Kamalika Ghosh, Co-PI: Debashis Chatterjee, Dept. of Electrical Engineering

- Worked on a SEPIC converter interfacing battery and PV cells where duty is controlled via Perturb & Observe (P&O) MPPT method. Along with that a buck converter to interface the battery and UV-C leds.

Past Projects 3: Small Signal Stability Analysis of Power System

Research Intern at Indian Institute of Technology Delhi

May'18 - July'18

Supervisor: Ashu Verma, Dept. of Energy Science and Engineering

- Analyzed small-signal stability of multi-bus multi-machine power system writing code in *Matlab*.

Academic Experince

0.1 Selected Teaching Assistantships

- at IIT KGP

M.Tech. Power Electronics Lab (**EE69247**) for the year of 2022-23-24-25, M. Tech. Machines Lab. (**EE69249**) for the year 2023 & 2025, 2nd Yr UG Electrical Machines Tutorial (**EE21206**) for the year 2024 & 2025, 2nd Yr UG Electrical Machines Tutorial (**EE29206**) for the year 2023-24- 25.

- at IIT Delhi

1st Yr UG Introduction to Electrical Engineering for the year 2021 (**ELL100**)

0.2 Conference Reviewing

IEEE ECCE-Asia 2025, IEEE INTELEC 2024

References

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Gautam Poddar

Professor, Indian Institute of Technology Kharagpur

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Subrata Saha

Chief Technologist , Aisin corporation, Aichi, Japan

subrata.saha@aisin.co.jp

Debashis Chatterjee

Professor, Jadavpur University

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